- Basic Commands 👍

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1. To check os version :
    - Switch to root → su -
    - Password : give root password
    - # cat /etc/redhat-release : for redhat operating system
    - # cat /etc/os-release : for all linux os (redhat+non redhat)
2. To check IP Address :
    - $ ip address show
    - $ ifconfig (not available in CLI Installation, switch to root and install net-tools package , then run)

```
[root@node11 ~]# dnf install net-tools
```

3. date : to see and change date and time

```
[student@node1 ~]$ date
Tue Oct  8 11:07:32 AM IST 2024
[student@node1 ~]$
[student@node1 ~]$ su -
Password:
Last login: Tue Oct  8 10:55:10 IST 2024 on pts/0
[root@node1 ~]#
[root@node1 ~]# man date
[root@node1 ~]#
[root@node1 ~]# # [MMDDhhmm[[CC]YY][.ss]]
[root@node1 ~]#
[root@node1 ~]# date 122512082026.33
Fri Dec 25 12:08:33 PM IST 2026
[root@node1 ~]#
```

```
[root@node1 ~]# # date -> month -> date -> hour -> minutes -> year -> .seconds
[root@node1 ~]# date 112620082029.44
Mon Nov 26 08:08:44 PM IST 2029
[root@node1 ~]#
```

4. To check and change hostname :

```
[student@node1 ~]$
[student@node1 ~]$ hostname
node1.example.com
[student@node1 ~]$
[student@node1 ~]$ hostnamectl set-hostname node11.example.com
Could not set static hostname: Access denied
[student@node1 ~]$
[student@node1 ~]$ su -
Password:
Last login: Thu Oct  3 09:51:08 IST 2024 on pts/0
[root@node1 ~]#
[root@node1 ~]# hostnamectl set-hostname node11.example.com
[root@node1 ~]#
[root@node1 ~]# hostname
node11.example.com
[root@node1 ~]#
```

5. who - shows all users who are accessing our server.

```
[student@node1 ~]$ who
root     tty1         2024-10-05 09:47
student  pts/0        2024-10-05 09:48 (172.19.10.78)
[student@node1 ~]$
[student@node1 ~]$
```

6. whoami - displays currently logged in users in the current terminal.

```
[student@node1 ~]$ whoami
student
[student@node1 ~]$
```

7. Why we are running this commands as we can already see the details in the prompt
    - Redhat has modified the prompt for us to understand , actually it shows the following way :
        - Sh-5.1$
8. Clear - to clear the screen

9. cal : to view calender

```
[root@node1 ~]# cal
    November 2029
Su Mo Tu We Th Fr Sa
                1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30
```

```
[root@node1 ~]# cal 04 2024
     April 2024
Su Mo Tu We Th Fr Sa
    1  2  3  4  5  6
 7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30
```

10. man : full form manual page. It shows the detailed information about a command. We need to press “q” to exit from the manual page.

```
[student@node1 ~]$ man fdisk
[student@node1 ~]$
[student@node1 ~]$
```

11. whereis : it shows both program path and execution path.

```
[student@node1 ~]$ whereis fdisk
fdisk: /usr/sbin/fdisk /usr/share/man/man8/fdisk.8.gz
```

12. which : it shows only execution path

```
[student@node1 ~]$ which mkdir
/usr/bin/mkdir
[student@node1 ~]$
```

13. help : this command show all available options that can be attached to a command

```
[student@node1 ~]$ mkdir --help
Usage: mkdir [OPTION]... DIRECTORY...
Create the DIRECTORY(ies), if they do not already exist.

Mandatory arguments to long options are mandatory for short options too.
  -m, --mode=MODE   set file mode (as in chmod), not a=rwx - umask
  -p, --parents     no error if existing, make parent directories as needed
  -v, --verbose     print a message for each created directory
  -Z                   set SELinux security context of each created directory
                         to the default type
      --context[=CTX]  like -Z, or if CTX is specified then set the SELinux
                         or SMACK security context to CTX
      --help     display this help and exit
      --version  output version information and exit

GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Full documentation <https://www.gnu.org/software/coreutils/mkdir>
or available locally via: info '(coreutils) mkdir invocation'
[student@node1 ~]$
```

14. info : it shows some glimpse information about a command

```
[student@node1 ~]$ info mkdir
[student@node1 ~]$
```

15. **Paths in Linux :-**

1. **Absolute Path :-**
    a. Starts from / [/ is the top directory in linux]
    b. This is the complete path of a content
    c. Searching is slower from this location
    d. Path can not exceed 4095 characters.
    e. Each filename can not exceed 255 characters.
    f. File names are case sensitive.

2. **Relative Path :-**
    a. Starts from the current location of the user
    b. It is the incomplete path of a content
    c. Searching is faster from this location

16. pwd : it shows the current working location. [pwd : Presently/Print Working Directory]

```
[student@node1 ~]$ pwd
/home/student
[student@node1 ~]$
```

17. uname : it shows All system details
    a. uname : display system information
    b. uname -a : display entire system's full information
    c. uname -s : kernel information
    d. uname -n : display system name [default→ localhost.localdomain]
    e. uname -r : kernel release information
    f. uname -m : system architecture
    g. uname -p : processor architecture
    h. uname -o : display operating system

18. mkdir : it is used to create a directory.

```
[student@node1 ~]$ mkdir database
```

19. To create a branch structure

```
[student@node1 ~]$ mkdir -p linux/{server,client}
```

20. To see the branch structure

```
[student@node1 ~]$ tree linux
linux
├── client
└── server
```

21. Another example of branch structure

```
[student@node1 ~]$ mkdir -p windows/win10/{bitlocker,firewall}
[student@node1 ~]$ tree windows
windows
├── win10
│   ├── bitlocker
│   └── firewall
├── win11
├── win7
└── win8.1
```

22. cd : change directory : it is used to navigate between directories

a.cd <location> : [rbs@station1 ~]$ cd linux

b.**cd ~** : to jump back to the home directory from any location

c.cd - : switch back to previous location :

d.**cd ..** : one “..” → one folder backward :

```
[student@node1 ~]$ cd windows
[student@node1 windows]$ cd
[student@node1 ~]$
[student@node1 ~]$ cd windows/win10
[student@node1 win10]$ cd
[student@node1 ~]$
[student@node1 ~]$ cd windows/win10/firewall/inbound/rule/portno/80/
[student@node1 80]$ cd
[student@node1 ~]$
[student@node1 ~]$ cd windows/win10/firewall/inbound/rule/portno/80/
[student@node1 80]$
[student@node1 80]$ cd ..
[student@node1 portno]$
[student@node1 portno]$ cd ../..
[student@node1 inbound]$
[student@node1 inbound]$ cd ../../..
[student@node1 windows]$
[student@node1 windows]$ cd -
/home/student/windows/win10/firewall/inbound
[student@node1 inbound]$
[student@node1 inbound]$ cd -
/home/student/windows
[student@node1 windows]$ cd ~
```

23. **File Creation :**

a. touch : it creates a blank file

```
[student@node1 ~]$ touch windows/win11/doc1.txt
[student@node1 ~]$ ls windows/win11/
doc1.txt
[student@node1 ~]$
[student@node1 ~]$ touch windows/win11/doc2
```

b. cat : it is used to create , view , add content inside a file

```
[student@node1 ~]$ cat > database/file1
Aaj Panchami , Aaj theke Biriyani Start.
[student@node1 ~]$
[student@node1 ~]$ cat database/file1
Aaj Panchami , Aaj theke Biriyani Start.
[student@node1 ~]$
[student@node1 ~]$ cat >> database/file1
Anek din wait kore acchi jekhane murgi dekhbo mere felbo
[student@node1 ~]$
[student@node1 ~]$ cat database/file1
Aaj Panchami , Aaj theke Biriyani Start.
Anek din wait kore acchi jekhane murgi dekhbo mere felbo
[student@node1 ~]$
[student@node1 ~]$ cat -a database/file1
cat: invalid option -- 'a'
Try 'cat --help' for more information.
[student@node1 ~]$ cat -b database/file1
     1  Aaj Panchami , Aaj theke Biriyani Start.
     2  Anek din wait kore acchi jekhane murgi dekhbo mere felbo
[student@node1 ~]$
```

c. gedit : it is also used to create a file. It is the same as Windows Notepad. But it is not installed by default in minimal installation.

```
[student@srv2 ~]$ gedit file2
[student@srv2 ~]$
[student@srv2 ~]$ cat file2
Saptami te 10 rokom macch khabo

[student@srv2 ~]$
```

d. nano : it is a text editor . We can create , modify , change the content of a file.

```
[student@node1 ~]$ nano database/file2
[student@node1 ~]$
[student@node1 ~]$ cat database/file2
Kaal shosthi , Kal ker din ta niramis
porer din bujhe nebo

[student@node1 ~]$
```

24. Calculator : bc : bring Calculator

```
[student@srv2 ~]$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free
 Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
25*25
625
^C
(interrupt) Exiting bc.
[student@srv2 ~]$
```

25. ls : It is used to view the contents of a directory.

```
[student@node1 tuesday15]$ ls
doc1  doc2
[student@node1 tuesday15]$
```

26. ls -l : to see the details of a content

```
[student@node1 ~]$ ls -l tuesday15/
total 0
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc1
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc2
[student@node1 ~]$
```

27. ls -ld : to see details of a folder / directory

```
[student@node1 ~]$ ls -ld tuesday15/
drwxr-xr-x. 2 student student 30 Oct 15 09:45 tuesday15/
[student@node1 ~]$
[student@node1 ~]$
```

28. ls -lh : h- human readable language

```
[root@node1 ~]# ls -lh /boot/vmlinuz-5.14.0-162.6.1.el9_1.x86_64
-rwxr-xr-x. 1 root root 12M Sep 30  2022 /boot/vmlinuz-5.14.0-162.6.1.el9_1.x86_64
[root@node1 ~]#
```

29. ls -li : i-inode number

An inode number in Linux is **a unique identifier for a file or directory on a Linux system**:

### What it is

An inode, or index node, is a file data structure that stores metadata about a file or directory. Each file or directory is assigned an inode number when it's created.

### What it contains

Inodes store information about a file or directory, including:

- File size
- File type
- Owner of the file
- Date and time of last access or modification
- Permissions and access control
- Physical location of the file on the disk

```
[student@node1 ~]$ ls -li tuesday15/doc1
174052 -rw-r--r--. 1 student student 0 Oct 15 09:45 tuesday15/doc1
[student@node1 ~]$
[student@node1 ~]$ ls -li tuesday15/doc1 tuesday15/doc2
174052 -rw-r--r--. 1 student student 0 Oct 15 09:45 tuesday15/doc1
174053 -rw-r--r--. 1 student student 0 Oct 15 09:45 tuesday15/doc2
[student@node1 ~]$
[student@node1 ~]$ ls -i tuesday15/doc1
174052 tuesday15/doc1
[student@node1 ~]$
[student@node1 ~]$
```

30. ls -a : a- all including hidden files and directories
31. To hide a file :

```
[student@node1 ~]$ ls -l tuesday15/
total 0
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc1
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc2
[student@node1 ~]$
[student@node1 ~]$ # I want to hide doc1
[student@node1 ~]$
[student@node1 ~]$ mv tuesday15/doc1 tuesday15/.doc1
[student@node1 ~]$
[student@node1 ~]$ ls -l tuesday15/
total 0
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc2
[student@node1 ~]$
```

32. To unhide a file :

```
[student@node1 ~]$ ls -la tuesday15/
total 0
drwxr-xr-x. 2 student student  31 Oct 15 10:50 .
drwx------. 6 student student 160 Oct 15 09:45 ..
-rw-r--r--. 1 student student   0 Oct 15 09:45 .doc1
-rw-r--r--. 1 student student   0 Oct 15 09:45 doc2
[student@node1 ~]$
[student@node1 ~]$ mv tuesday15/.doc1 tuesday15/doc1
[student@node1 ~]$ ls -l tuesday15/
total 0
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc1
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc2
```

33. ll : alternative command for ls -l :- shows long list

```
[student@node1 ~]$ ll
total 0
drwxr-xr-x. 2 student student 32 Oct  8 10:56 database
drwxr-xr-x. 4 student student 34 Oct  8 10:09 linux
drwxr-xr-x. 2 student student 30 Oct 15 10:51 tuesday15
drwxr-xr-x. 6 student student 58 Oct  8 10:12 windows
[student@node1 ~]$
[student@node1 ~]$ ll tuesday15/
total 0
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc1
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc2
[student@node1 ~]$
```

34. cp : It is used to copy and paste a content

```
[student@node1 ~]$ ls -l tuesday15/
total 0
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc1
-rw-r--r--. 1 student student 0 Oct 15 09:45 doc2
[student@node1 ~]$
[student@node1 ~]$ ls -l wednesday16/
total 0
[student@node1 ~]$
[student@node1 ~]$ cp -r tuesday15/doc1 wednesday16/
[student@node1 ~]$
[student@node1 ~]$ ls -l wednesday16/
total 0
-rw-r--r--. 1 student student 0 Oct 15 11:03 doc1
[student@node1 ~]$
```

35. Copy from multiple location and paste in a single location

```
[student@node1 ~]$ cp -r monday14/file1 tuesday15/file2 wednesday16/{file3,file4} thursday17/file5 fri
day18/
[student@node1 ~]$ ls -l friday18/
total 0
-rw-r--r--. 1 student student 0 Oct 15 11:10 file1
-rw-r--r--. 1 student student 0 Oct 15 11:10 file2
-rw-r--r--. 1 student student 0 Oct 15 11:10 file3
-rw-r--r--. 1 student student 0 Oct 15 11:10 file4
-rw-r--r--. 1 student student 0 Oct 15 11:10 file5
[student@node1 ~]$
```

36. Copy all from a location : v : verbose : view the background process

```
[student@node1 ~]$ cp -r friday18/* saturday19/
[student@node1 ~]$
[student@node1 ~]$ mkdir sunday20
[student@node1 ~]$
[student@node1 ~]$ cp -rv friday18/* sunday20/
'friday18/file1' -> 'sunday20/file1'
'friday18/file2' -> 'sunday20/file2'
'friday18/file3' -> 'sunday20/file3'
'friday18/file4' -> 'sunday20/file4'
'friday18/file5' -> 'sunday20/file5'
[student@node1 ~]$
[student@node1 ~]$ ls -l sunday20/
total 0
-rw-r--r--. 1 student student 0 Oct 15 11:14 file1
-rw-r--r--. 1 student student 0 Oct 15 11:14 file2
-rw-r--r--. 1 student student 0 Oct 15 11:14 file3
-rw-r--r--. 1 student student 0 Oct 15 11:14 file4
-rw-r--r--. 1 student student 0 Oct 15 11:14 file5
[student@node1 ~]$
```

37. echo : to display a file content , also we can insert text to a file.

```
[student@node1 thursday]$ touch doc1
[student@node1 thursday]$ echo "Ishan khub bhalo chele" > doc1
[student@node1 thursday]$
[student@node1 thursday]$ cat doc1
Ishan khub bhalo chele
[student@node1 thursday]$
```

38. mv : full form is move. It is used to cut, paste , rename , hide and unhide.

a.Cut & paste

```
[student@node1 ~]$ mkdir data1
[student@node1 ~]$ ls -l thursday
total 4
-rw-r--r--. 1 student student 23 Oct 17 09:47 doc1
[student@node1 ~]$
[student@node1 ~]$ ls -l data1
total 0
[student@node1 ~]$
[student@node1 ~]$ mv thursday/doc1 data1/
[student@node1 ~]$
[student@node1 ~]$ ls -l data1
total 4
-rw-r--r--. 1 student student 23 Oct 17 09:47 doc1
[student@node1 ~]$
[student@node1 ~]$ ls -l thursday
total 0
[student@node1 ~]$
```

b.Rename :

```
[student@node1 ~]$ ls data1
doc1
[student@node1 ~]$ mv data1/doc1 data1/document-1
[student@node1 ~]$
[student@node1 ~]$ ls data1
document-1
[student@node1 ~]$
```

c.Hide and unhide

```
[student@node1 ~]$ ls data1
document-1
[student@node1 ~]$
[student@node1 ~]$ mv data1/document-1 data1/.document-1
[student@node1 ~]$
[student@node1 ~]$ ls data1
[student@node1 ~]$
[student@node1 ~]$ ls -a data1
.   ..   .document-1
[student@node1 ~]$
[student@node1 ~]$ mv data1/.document-1 data1/document-1
[student@node1 ~]$
[student@node1 ~]$ ls data1
document-1
[student@node1 ~]$
```

39. rm and rmdir : rm means remove. Rmdir means removing a directory.

```
[student@node1 ~]$ mkdir akash
[student@node1 ~]$ ls
akash   data1     friday18  linux     saturday19  thursday    tuesday15    windows
behala  database  ishan     monday14  sunday20    thursday17  wednesday16
[student@node1 ~]$
[student@node1 ~]$ rmdir akash
[student@node1 ~]$
[student@node1 ~]$ ls
behala  database  ishan  monday14    sunday20  thursday17  wednesday16
data1   friday18  linux  saturday19  thursday  tuesday15   windows
[student@node1 ~]$
```

40. rmdir can only delete empty directories **

```
[student@node1 ~]$ rm data1/document-1
[student@node1 ~]$
[student@node1 ~]$ ls data1/
[student@node1 ~]$
[student@node1 ~]$
```

41. For forceful deletion

```
[student@node1 ~]$ rm -rf database/
[student@node1 ~]$
[student@node1 ~]$ ls
behala    ishan  monday14    sunday20  thursday17  wednesday16
friday18  linux  saturday19  thursday  tuesday15   windows
[student@node1 ~]$
```

42. Deleting multiple files

```
[student@node1 ~]$ ls sunday20/
file1  file2  file3  file4  file5
[student@node1 ~]$
[student@node1 ~]$
[student@node1 ~]$ rm sunday20/file{1..3}
[student@node1 ~]$ ls sunday20/
file4  file5
[student@node1 ~]$
[student@node1 ~]$ rm sunday20/{file4,file5}
[student@node1 ~]$
```

43. head : it is used to see a custom number of lines from the beginning. By default it will show 10 lines but we can customize the count

```
[student@node1 ~]$ head /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
[student@node1 ~]$
```

```
[student@node1 ~]$ head -n 3 /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
[student@node1 ~]$
```

44. tail : it is used to view lines of a file from bottom . it also shows 10 lines by default but we can customize the count

```
[student@node1 ~]$ tail /etc/passwd
dbus:x:81:81:System message bus:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/dev/null:/sbin/nologin
sssd:x:998:995:User for sssd:/:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:997:994::/var/lib/chrony:/sbin/nologin
systemd-oom:x:992:992:systemd Userspace OOM Killer:/:/usr/sbin/nologin
student:x:1000:1000::/home/student:/bin/bash
devops:x:1001:1001::/home/devops:/bin/bash
pronoy:x:1002:1002::/home/pronoy:/bin/bash
ali:x:1003:1003::/home/ali:/bin/bash
```

```
[student@node1 ~]$ tail -n 4 /etc/passwd
student:x:1000:1000::/home/student:/bin/bash
devops:x:1001:1001::/home/devops:/bin/bash
pronoy:x:1002:1002::/home/pronoy:/bin/bash
ali:x:1003:1003::/home/ali:/bin/bash
[student@node1 ~]$
```

45. To find out a file type
    a. $ file <filename>
46. To see already run commands
    a. $ history
47. To run a command from history
    a. $ !line number from history

48. To start reverse search from history
    a. Run command from history searching from below
        i. $ !<command>